NRF5340DK Flashing and Tesing Guide

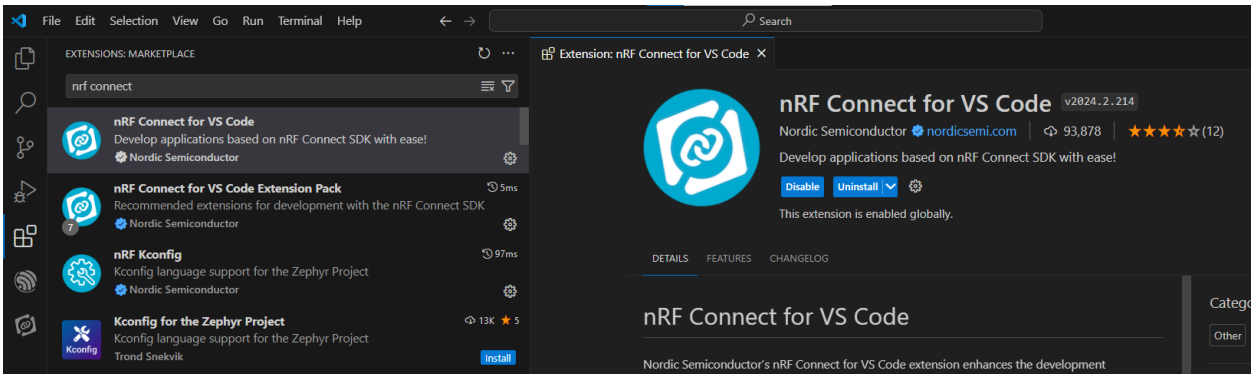
1 - Setup NRF connect SDK on VS Code

A. VS Code Installation

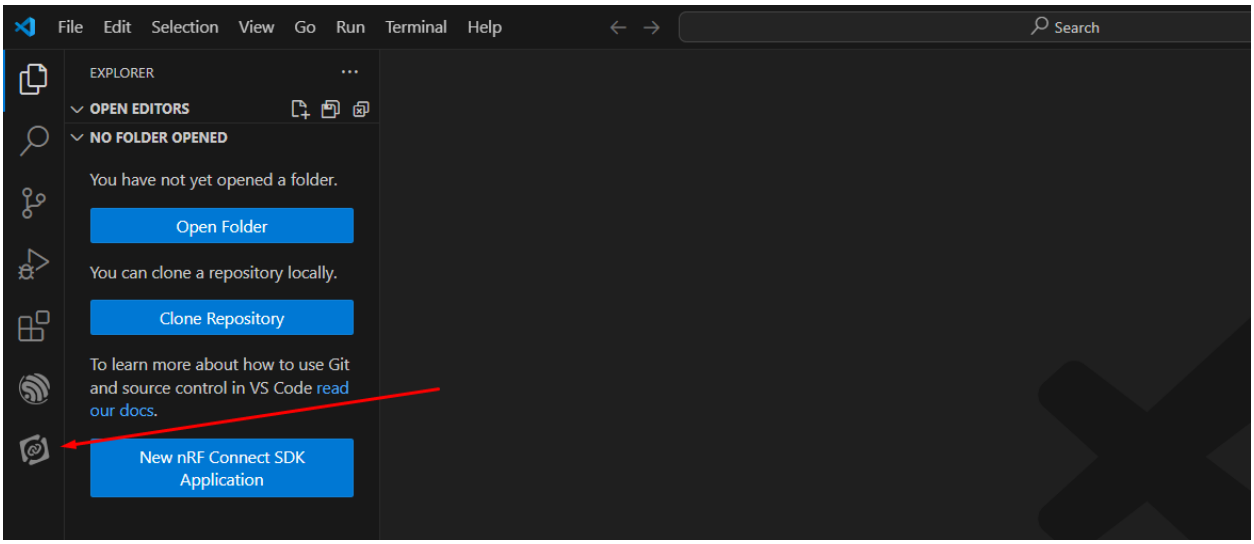
1. Download the Visual Studio Code installer for Windows.
2. Once it is downloaded, run the installer (VSCodeUserSetup-{version}.exe). This will only take a minute.
3. Open VS Code.

B. Add NRF Connect SDK extension in VS Code.

1. Add both the nrf connect sdk extensions.

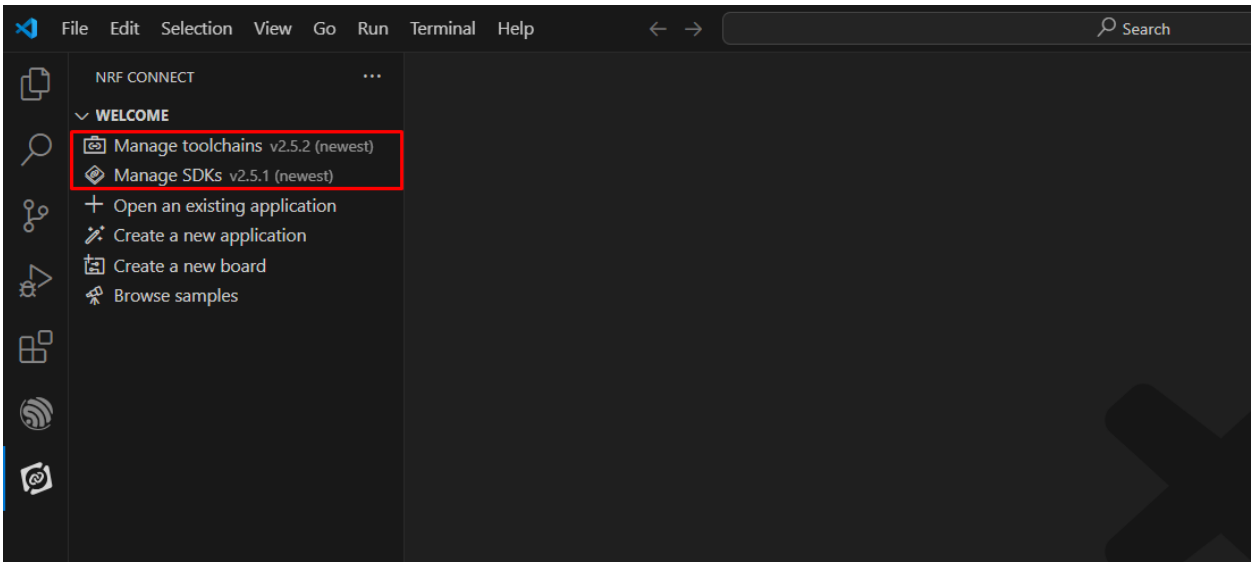


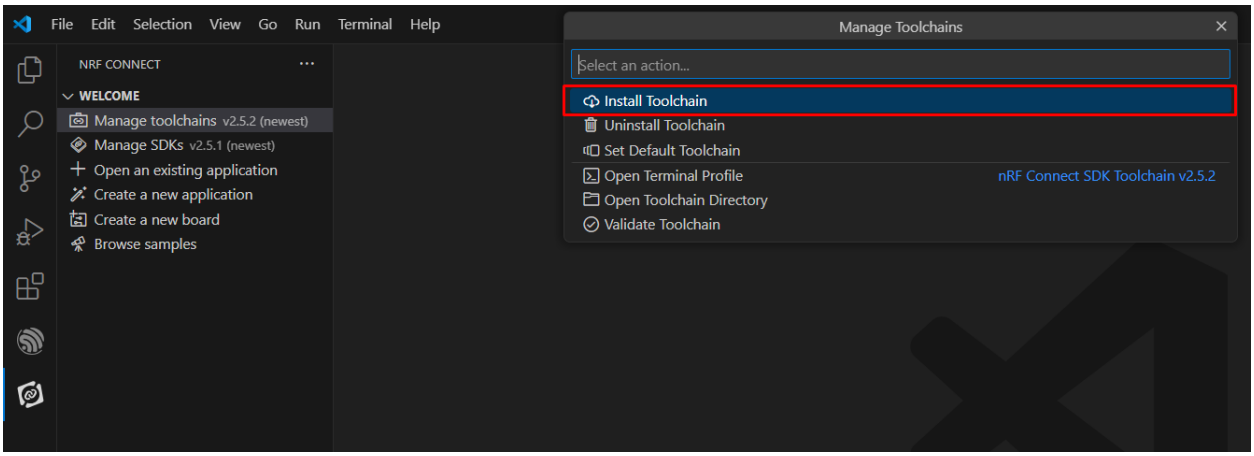
2. After installing the extensions nrf connect extension icon will appear on the left bar.

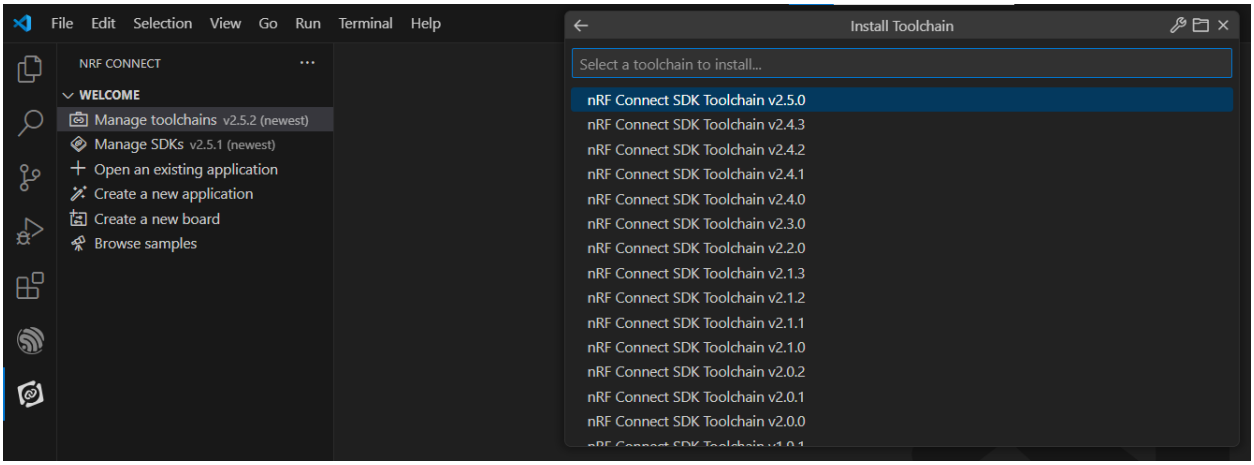


- Install Toolchain

Click on Manage toolchain, then install the nRF Connect SDK Toolchain v2.7.0.

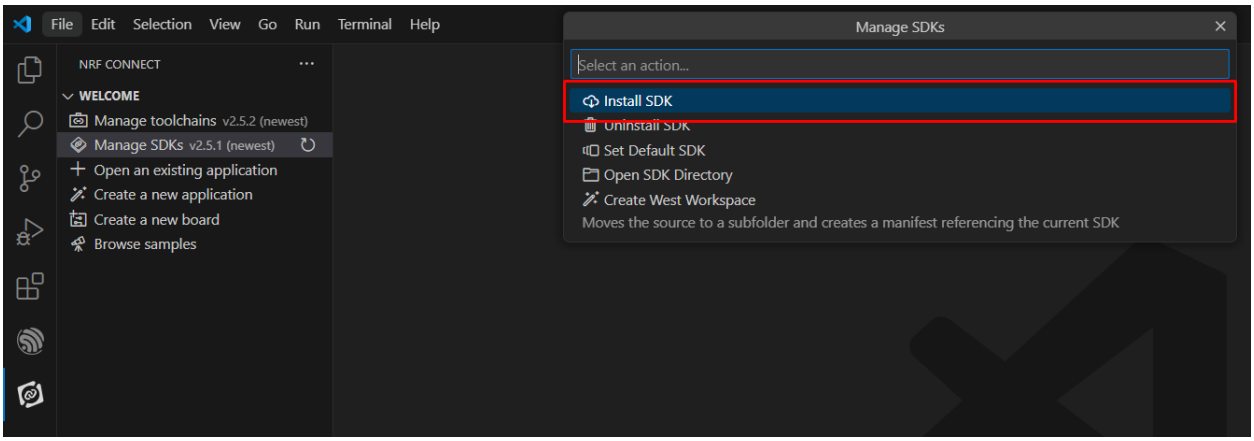


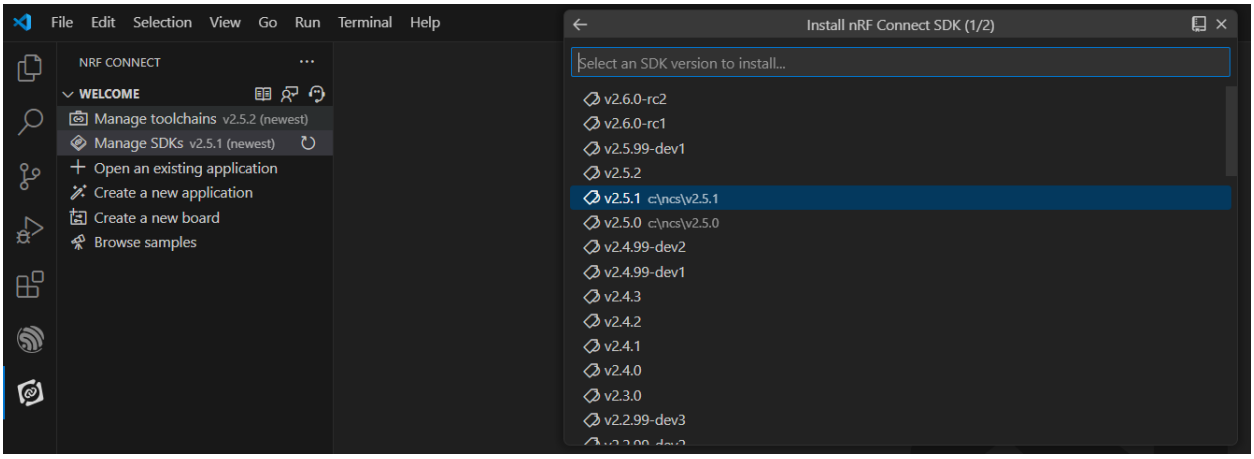




- Install SDK

Then do the same for Manage SDKs, install v2.7.0





2 - Install J-Link tools

Download J-Link tools from here We used v7.94e. Download the latest from this link for the tools here

- Follow the installation guide and install.


SEGGER - J-Link V7.96 (64-bit Windows) Setup
Embedded Studio
Powerful C/C++ IDE available for
Windows
macOS
Linux
Download Trial
SEGGER
Welcome to SEGGER - J-Link V7.96 (64-bit Windows) Setup
Setup will guide you through the installation of SEGGER - J-Link V7.96 (64-bit Windows).
It is recommended that you close all other applications before starting Setup. This will make it possible to update relevant system files without having to reboot your computer.
Click Next to continue.
Next >
Cancel

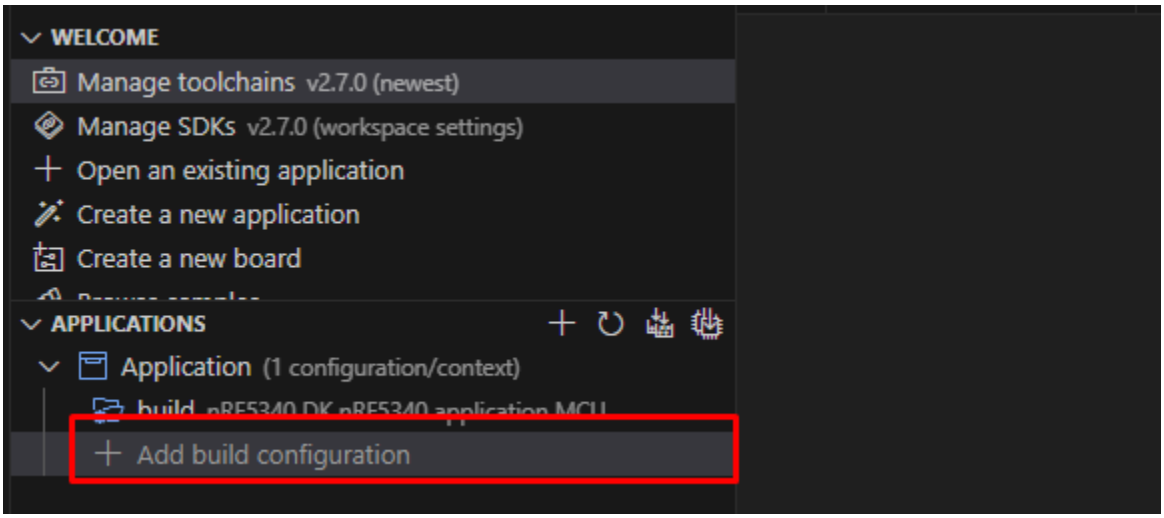
Connecting Peripherals

1. **RGB LED Strip**:
    a. Connect the **control pin** to P1.13 on the nRF5340-DK.
    b. Power the strip separately if it requires a higher current than the DK can supply.
    c. Common ground must be shared between the strip and the DK.

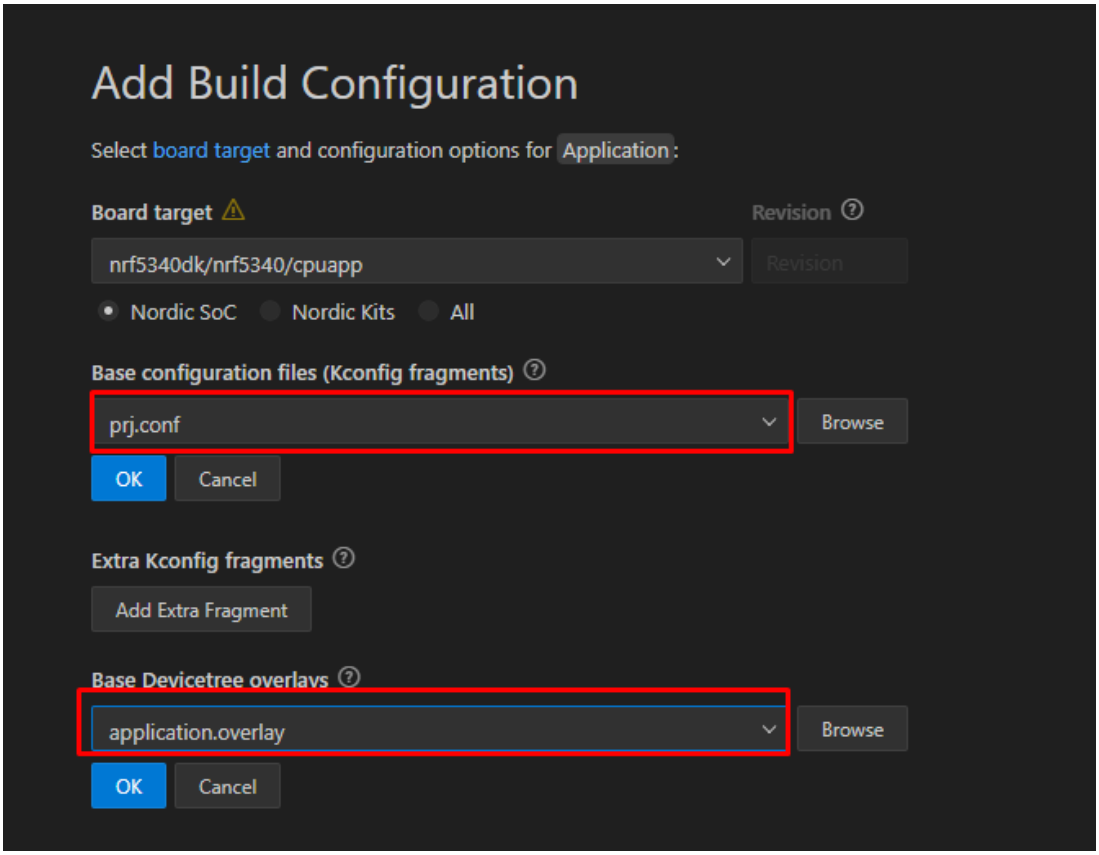
2. **INMP441 Microphone**:
    a. Use the I2S interface pins:
        i. WS: P1.12.
        ii. CLK: P1.15.
        iii. SD: 1.14.
        iv. LR: GND
        v. VDD: 3V3
        vi. GND: GND
    b. Ensure power (VDD) and ground are connected properly.
3. **Audio Amplifier**:
    a. Connect the amplifier's I2S inputs to the same I2S lines as the microphone. SD Pin will be different.
        i. LRCLK: P1.12.
        ii. BCLK: P1.15.
        iii. IIS_DATA: 1.14.
        iv. GND: GND
    b. Ensure the audio amplifier is separately powered and GND is common to NRF5340DK.

Flashing the Device

1. Go to NRF Connect Extention Tab. Select “Add Build Configuration”.

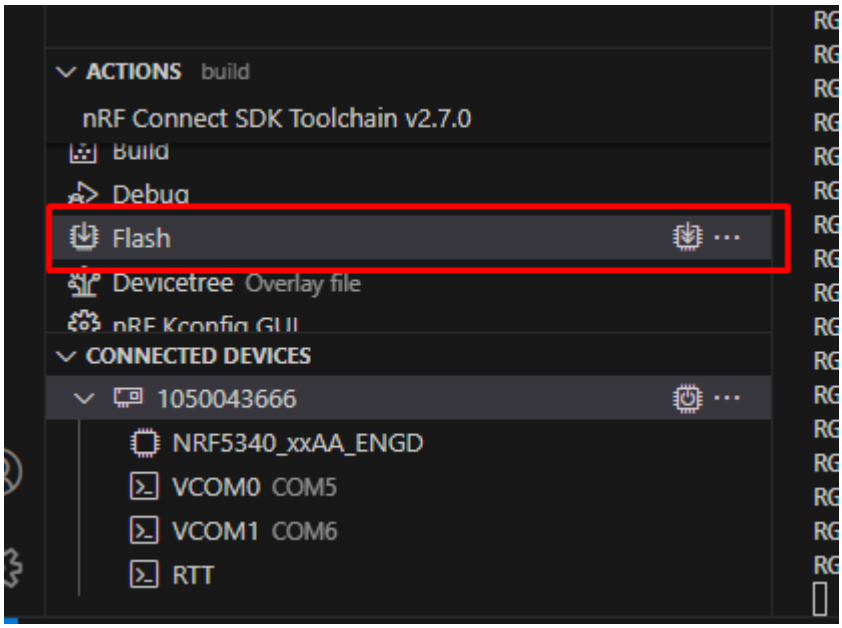


2. Select “prj.conf” as Base configuration files and “application.overlay” as Base Devicetree overlay.



3. Click Build Configuration.

4. Once build, the device can be flashed from the Actions section.



Behaviour

1. The speaker should stream the real-time audio from Microphone.
2. The RGB will be flashing random colors.